if

```
x = 6                              Blok 1
if x > 5:
    x = x - 1                      Blok 2
    if x == 5:
        print("Jubii")             Blok 3
    x = x + 1                      Blok 2 - fortsat
print(x)                           Blok 1 - fortsat
```

Funktioner

The function song

```
print("Functions are super, Functions are cool")
print("When writing a program they are a great tool")
print("La la dim du da da di")
print("Skubi dubi dumdi di")
print("Bing di dubi dum da di")

print("Functions are useful to package some code")
print("They are not so strange that your head will explode")
print("La la dim du da da di")
print("Skubi dubi dumdi di")
print("Bing di dubi dum da di")
```

```python
def chorus():
    print("La la dim du da da di")
    print("Skubi dubi dumdi di")
    print("Bing di dubi dum da di")

print("Functions are super, Functions are cool")
print("When writing a program they are a great tool")
chorus()

print("Functions are useful to package some code")
print("They are not so strange that your head will explode")
chorus()
```

Definition af en funktion

```
def chorus():
    print("La la dim du da da di")
    print("Skubi dubi dumdi di")
    print("Bing di dubi dum da di")
```

```python
def chorus():
    print("La la dim du da da di")
    print("Skubi dubi dumdi di")
    print("Bing di dubi dum da di")

print("Functions are super, Functions are cool")
print("When writing a program they are a great tool")
chorus()

print("Functions are useful to package some code")
print("They are not so strange that your head will explode")
chorus()
```

Returning values

```python
def f():
    return 2
```

Expressions med funktionskald

```
def f():
    return 2


print(f() * 8)
```

```
         2 >= 8
```

substitution

```
          False
```

reduktion

Tanketjek

```python
def f():
    return 2

print(f() + f() + f())
print(f() ** 8)
print(f)
```

Hello Mogens

Skriv en funktion der returnerer værdien:

`"Hello Mogens"`

Greeting

```
def greeting():
    name = "Mogens"
    return "Hello " + name

message = greeting()
print(message)
```

Værdi-input

```
# def greeting():
#       name = "Mogens"
#       return "Hello " + name

def greeting(name):
    return "Hello " + name

message = greeting("Mogens")
print(message)
```

Værdi-input

Variabel (parameter)

```
def greeting(name):
    return "Hello " + name
```

Værdi (argument)

```
message = greeting("Mogens")
print(message)
```

Værdi-input

```
def greeting(name):
    return "Hello " + name


message = greeting("Mog" + "ens")
print(message)
```

Værdi-input

```
def greeting(name):
    return "Hello " + name

x = "Mogens"
message = greeting(x)
print(message)
```

Miniopgave

Skriv en funktion der returnerer kvadratet af et tal:

`square(4)`

(skal returnere 16)

Flere argumenter

```
def division(n, d):
    result = n / d
    return result

fraction = division(3, 10)
print(fraction)
```

Miniopgave

Skriv en funktion der returnerer summen af to tal:

```
summation(4, 3)
```

(skal returnere 7)

Funktioner

“*Temporary* mini-world”
(Call frame)

```
def greeting(name):
    salute = "Hello"
    return salute + " " + name

message = greeting("Mogens")
print(message)
```

Miniopgave

Skriv en funktion der returnerer kvadratet af et tal:

`square(4)`

(skal returnere 16)

Miniopgave

Skriv en funktion der returnerer summen af kvadrater (sum of squares)

```
sum_of_squares(4, -2, 3)
```

(skal returnere 29)

Løsning

Løsning

Funktionenskaldets midlertidige og private verden

"*Temporary* mini-world"

```
def greeting(name):
    salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
print(message)
```

Funktionens private midlertidige verden

```
def greeting(name):
    salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
print(message)
```

`salute` is private to the function

Funktionens private midlertidige verden

```
def greeting(name):
    salute = "Hello"
    return salute + " " + name

salute = "Bonjour"
message = greeting("Mogens")
print(message)
print(salute)
```

`salute` is *still* private to the function

Funktionens private midlertidige verden

Python has to look for **`salute`** *outside* the function. Does it find it?

```
def greeting(name):
    # salute = "Hello"
    return salute + " " + name

salute = "Bonjour"
message = greeting("Mogens")

print(message)
```

Funktionens private midlertidige verden

Python has to look for **`salute`** *outside* the function. Does it find it?

```
def greeting(name):
    # salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
salute = "Bonjour"
print(message)
```

Variable og funktioner

- Et funktionskald skaber en midlertidig verden som forsvinder igen når funktionen returnerer en værdi.
- Alle variable man giver værdier i en funktion er private for funktionen.
- Hvis en variabel ikke er defineret i en funktion vil Python prøve at finde en definition udenfor funktionen.

Hvad er galt?

```
def greeting(name):
    name = ""
    salute = "Hello"
    return salute + " " + name


message = greeting("Mogens")
print(message)
```

Hvad er galt?

```
def greeting(nama):
    salute = "Hello"
    return salute + " " + name

name = "Henning"
message = greeting("Mogens")
print(message)
```

Builtin functions

```
abs_val = abs(-42)
print(abs_val)

fruit = "Banana"
length = len(fruit)
print(length)

print(type(length))
print(type(fruit))
```

https://docs.python.org/3.8/library/functions.html

Python 3.9 dokumentation

https://docs.python.org/3.9/

Look for "Library Reference"

Miniopgave

Skriv en funktion der tester om et tal er lige eller ulige

`is_even(42)`

skal returnere `True`

`is_even(17)`

skal returnere `False`

Løsning